A1.1 About the AXI protocol

The AMBA AXI protocol supports high-performance, high-frequency system designs.

The AXI protocol:

- is suitable for high-bandwidth and low-latency designs
- provides high-frequency operation without using complex bridges
- meets the interface requirements of a wide range of components
- is suitable for memory controllers with high initial access latency
- provides flexibility in the implementation of interconnect architectures
- is backward-compatible with existing AHB and APB interfaces.

The key features of the AXI protocol are:

- separate address/control and data phases
- support for unaligned data transfers, using byte strobes
- uses burst-based transactions with only the start address issued
- separate read and write data channels, that can provide low-cost *Direct Memory Access* (DMA)
- support for issuing multiple outstanding addresses
- support for out-of-order transaction completion
- permits easy addition of register stages to provide timing closure.

The AXI protocol includes the optional extensions that cover signaling for low-power operation.

The AXI protocol includes the AXI4-Lite specification, a subset of AXI4 for communication with simpler control register style interfaces within components. See Chapter B1 *AMBA AXI4-Lite*.

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